

Lab 10

EGRE246

This week you will get some experience working with the Stack ADT.

1. Please follow the lecture of Stack structure to make stack.h and stack.c and use the following code to test you Stack ADT.

```
#include <stdio.h>
#include <stdlib.h>
#include <limits.h>
#include "stack.h"
int main(void) {
    stack s1 = createStack(),s2;
    printf("stack #1: ");
    printStack(s1);
    printf("\n");
    for(int i = 2; i <= 14; i+=2)
        push(s1,i);
    printf("stack #2: ");
    printStack(s1);
    printf("\n");
    printf("get(2) = %d\n",get(s1,2));
    printf("get(6) = %d\n",get(s1,6));
    printf("get(16) = %d\n",get(s1,16));
    printf("get(-1) = %d\n",get(s1,-1));
    printf("stack #3: ");
    printStack(s1);
    printf("\n");
    pop(s1);
    pop(s1);
    printf("stack #4: ");
    printStack(s1);
    printf("\n");
    put(s1,100,2);
    put(s1,200,99);
    printf("stack #5: ");
    printStack(s1);
    printf("\n");
    s2 = createStack();
    put(s2,1000,5);
    printf("stack #6: ");
    printStack(s2);
    printf("\n"); */
```

```
    return 0;  
}
```

2. In your Stack ADT, you need to implement the basic functions createStack, push, and pop functions, you also need to implement the following additional functions.

```
void printStack(stack);  
// prints the stack
```

```
int get(stack s,int p);  
// returns but does not remove the item in s at position p;  
// returns INT_MIN if size(s) == 0 or p < 0 or p >= size(s)
```

```
bool put(stack s,int x,int p);  
// inserts x in s at position p, returning true if item is  
// successfully inserted; return false if p < 0 or you  
// cannot insert item because stack is full;  
//when inserting x into the stack, you might need to move some items to make a slot for insertion  
//and adjust the top position.
```

3. You can use regular array or dynamic array to implement your Stack ADT.
4. Finishing up by coding and testing all the functions.
5. Run of completed project and submit your source code including .h and .c files on Canvas.